

Foundations of Sentiment analysis

[Sentiment analysis]

1.0 Content Block

Types A basic task in sentiment analysis is classifying the polarity of a given text at the document, sentence, or feature/aspect level—whether the expressed opinion in a document, a sentence or an entity feature/aspect is positive, negative, or neutral. Advanced, "beyond polarity" sentiment classification looks, for instance, at emotional states such as enjoyment, anger, disgust, sadness, fear, and surprise. Precursors to sentimental analysis include the General Inquirer, which provided hints toward quantifying patterns in text and, separately, psychological research that examined a person's psychological state based on analysis of their verbal behavior. Subsequently, the method described in a patent by Volcani and Fogel, looked specifically at sentiment and identified individual words and phrases in text with respect to different emotional scales. A current system based on their work, called EffectCheck, presents synonyms that can be used to increase or decrease the level of evoked emotion in each scale. Many other subsequent efforts were less sophisticated, using a mere polar view of sentiment, from positive to negative, such as work by Turney, and Pang who applied different methods for detecting the polarity of product reviews and movie reviews respectively. This work is at the document level. One can also classify a document's polarity on a multi-way scale, which was attempted by Pang and Snyder among others: Pang and Lee expanded the basic task of classifying a movie review as either positive or negative to predict star ratings on either a 3- or a 4-star scale, while Snyder performed an in-depth analysis of restaurant reviews, predicting ratings for various aspects of the given restaurant, such as the food and atmosphere (on a five-star scale). First steps to bringing together various approaches—learning, lexical, knowledge-based, etc.—were taken in the 2004 AAAI Spring Symposium where linguists, computer scientists, and other interested researchers first aligned interests and proposed shared tasks and benchmark data sets for the systematic computational research on affect, appeal, subjectivity, and sentiment in text. Even though in most statistical classification methods, the neutral class is ignored under the assumption that neutral texts lie near the boundary of the binary classifier, several researchers suggest that, as in every polarity problem, three categories must be identified. Moreover, it can be proven that specific classifiers such as the Max Entropy and SVMs can benefit from the introduction of a neutral class and improve the overall accuracy of the classification. There are in principle two ways for operating with a neutral class. Either, the algorithm proceeds by first identifying the neutral language, filtering it out and then assessing the rest in terms of positive and negative sentiments, or it builds a three-way classification in one step. This second approach often involves estimating a probability distribution over all categories (e.g. naive Bayes classifiers as implemented by the NLTK). Whether and how to use a neutral class depends on the nature of the data: if the data is clearly clustered into neutral, negative and positive language, it makes sense to filter the neutral language out and focus on the polarity between positive and negative sentiments. If, in contrast, the data are mostly neutral with small deviations towards positive and negative affect, this strategy would make it harder to clearly distinguish between the two poles. A different method for determining sentiment is the use of a scaling system whereby words commonly associated with having a negative, neutral, or positive sentiment are given an associated number on a -10 to +10 scale (most negative up to most positive) or simply from 0 to a positive upper limit such as +4. This makes it possible to adjust the sentiment of a given term relative to its environment (usually on the level of the sentence). When a piece of unstructured text is analyzed using natural language processing, each concept in the specified environment is given a score based on the way sentiment words relate to the concept and its associated score. This allows movement to a more sophisticated understanding of sentiment, because it is now possible to adjust the sentiment value of a concept relative to modifications that may surround it. Words, for example, that intensify, relax or negate the sentiment expressed by the concept can affect its score. Alternatively, texts can be given a positive and negative sentiment strength

score if the goal is to determine the sentiment in a text rather than the overall polarity and strength of the text. There are various other types of sentiment analysis, such as aspect-based sentiment analysis, grading sentiment analysis (positive, negative, neutral), multilingual sentiment analysis and detection of emotions.

2.0 Content Block

Metaphorical expressions. The text contains metaphoric expression may impact on the performance on the extraction. Besides, metaphors take in different forms, which may have been contributed to the increase in detection. Discrepancies in writings. For the text obtained from the Internet, the discrepancies in the writing style of targeted text data involve distinct writing genres and styles. Context-sensitive. Classification may vary based on the subjectiveness or objectiveness of previous and following sentences. Time-sensitive attribute. The task is challenged by some textual data's time-sensitive attribute. If a group of researchers wants to confirm a piece of fact in the news, they need a longer time for cross-validation, than the news becomes outdated. Cue words with fewer usages. Ever-growing volume. The task is also challenged by the sheer volume of textual data. The textual data's ever-growing nature makes the task overwhelmingly difficult for the researchers to complete the task on time. Previously, the research mainly focused on document level classification. However, classifying a document level suffers less accuracy, as an article may have diverse types of expressions involved. Researching evidence suggests a set of news articles that are expected to dominate by the objective expression, whereas the results show that it consisted of over 40% of subjective expression. To overcome those challenges, researchers conclude that classifier efficacy depends on the precisions of patterns learner. And the learner feeds with large volumes of annotated training data outperformed those trained on less comprehensive subjective features. However, one of the main obstacles to executing this type of work is to generate a big dataset of annotated sentences manually. The manual annotation method has been less favored than automatic learning for three reasons: